AR-Mini · Lab protocol v2

Two desks (Claude, Cursor), two consultants (Gemini, ChatGPT), one human. AI proposes; Teddy builds, measures and decides.

AI proposes · the measurement decides

r1a 11.10 · r2a 16.10 · r3a 26.10 · r4 31.10 · submit 01.11

1 The day

what comes after what, in which tool

Morning — 10 min

GitHub: pull requests → green and clear = merge into dev. Board: at most 3 cards in Today. Telegram: any "order by ..."?

School — 0 min

The agents work on their own: night tasks in Cursor (night/*), the parts agent every hour on GitHub.

Afternoon · evening — cards

C### in order: hypothesis → Claude → test → photo → Close. One thing at a time.

Before bed — 10 min

Cursor: "commit: ...". Photos → 00_Inbox → "C006 inbox: ..." to Claude. Start a print or a night task.

2 How one card runs

one card = one hypothesis = one ID C###

1 Card (Board)

Board: Backlog → Today. Title + hypothesis with a number and a tolerance. Max 3 in Today.

2 Ask (Claude)

Workshop chat: "C006: what do I measure and how?" 3 lines of context + the question. Before 12 V or a long print → ChatGPT for a second opinion.

3 Prepare (lab)

Instruments, multimeter battery, iPhone on the stand, light. Safety: section 6.

4 Do (lab)

Print, assemble, wire, measure. Never change two things in one test.

5 Film (iPhone)

The instrument in frame with every number. Video 4K landscape, 5 s before and after.

6 Close (Board)

Result with the numbers, Pass / Partial / Fail, the next fix → Done.

7 Record (Cursor)

A file changed → "commit: ... · C006". Fail → FAILS_AND_FIXES. Photos → 00_Inbox → "C006 inbox: ..." to Claude.

A Fail is not a failure: it is the next version. A release is cut when there is something to show, not by date.

3 Which tool for what

two desks, two consultants, three places for files

Claude · workshop chat	Desk 1	decisions, CAD / EDA files, specs for the agents, logistics, portfolio, sorts iCloud
AR-Mini Board	Claude	cards C###, releases r2a..., Export JSON (backup in the repo)
Cursor · chat	Desk 2	commit: · inbox · add need: · ordered · received · parts report · fix:
Cursor · Background Agent	Desk 2	night tasks T#: one line → branch night/* → PR to dev
GitHub	truth	dev / main · PR · Actions: CI + hourly parts agent · Releases with the video
iCloud · AR-Mini-Media	media	00_Inbox → 01_Loops/C###_date_name · captions.md · never in git
Gemini	consultant	video: "at which second does the tilt start?" · datasheets
ChatGPT	consultant	second opinion before 12 V, LiPo, a long print: "what can go wrong?"
Telegram	alerts	parts agent: "order N003 by 14.10" · "PR green"

How I ask Claude

```
C006 · esc-9v
Context: thruster with the ESC
inside, 4 wires; 9 V from the
bench supply, limit 0.10 A.
Question: how do I know it arms?
I want: a number ±, how to
measure it, the pass criterion.
```

- One card = one question.
- Claude proposes. You decide and measure.
- If it says "it works" → "how did we measure it?"
- It writes files straight into the repo; you tell Cursor "commit:".

4 The rules

twelve, no exceptions

1. A hypothesis with a number and a tolerance before you touch anything. Without a number it is not a hypothesis.
2. One card = one thing. Never change two things in one test.
3. The number comes from an instrument, not the eye. The instrument is in frame.
4. Pass / Fail is about the design, not about you. Fail = the next version, written into FAILS_AND_FIXES.
5. At most 3 cards in Today. A new one goes in when one comes out.
6. AI proposes, you decide. Nothing is done "because the agent said so".
7. Every commit goes through Cursor and check.sh. AI never pushes to main. Only tested work goes into main.
8. A night task = a night/* branch and a PR. Merge only what you understand.
9. Photos and video → iCloud, never in git. The folder carries the card ID.
10. A release is cut when there is something to show. Release = Export JSON + commit + video.
11. A part is missing → "add need: ..." the same day. The agent searches, you order.
12. Passwords and keys only in Secrets / secrets.h. The manuscript never goes into a public repo. In the portfolio, only what was measured.

5 One ID everywhere

C006 ties everything together

Board	card C006 · ESC arms at 9 V
Chat	`C006 · esc-9v`
iCloud	`01_Loops/C006_2026-10-09_esc-9v/`
Files	`C006_01.HEIC … C006_07.MOV`
Captions	captions.md in the same folder
Commit	`fw: esc arm at 9 V · C006`
Video	caption "C006 · ESC arms at 9 V · Pass"
Release	r2a · 16.10

6 Safety

before every test

- Multimeter: continuity and shorts BEFORE power.
- 12 V only to a checked part. First from the bench supply with a current limit.
- LiPo: charge only in the bag and supervised.
- Smoke, smell, heat: switch off first, then take the photo.
- Soldering iron and fume extractor: switch them on together, switch the iron off first.